

Exact q -Binomial Extraction of Rvachev Up-Function Values from Finite Sinc-Product Splines

Research article prepared for the ProveIt/FabiusFunction project

2 September 2026

Abstract

Let

$$\widehat{\text{up}}(t) = \prod_{k=0}^{\infty} \text{sinc}\left(\frac{\pi t}{2^k}\right), \quad \text{sinc } z = \frac{\sin z}{z},$$

and let up_n be the inverse Fourier transform of the prefix product with $0 \leq k \leq n$. Fix a dyadic rational $x \in [-1, 1]$, and write $d = d(x) = \min\{r \geq 0 : 2^r x \in \mathbb{Z}\}$ and $M = \lfloor d/2 \rfloor$. The principal result is the exact eventual identity

$$\text{up}_n(x) = \sum_{m=0}^M (-1)^m a_m \text{up}^{(2m)}(x) 4^{-m(n+1)}, \quad n \geq d,$$

where the rational coefficients a_m are defined by

$$\frac{1}{\widehat{\text{up}}(z)} = \sum_{m \geq 0} a_m (2\pi z)^{2m}.$$

Thus the sequence $q_n = \text{up}_n(x)$ is, after at most d initial levels, exactly a constant plus finitely many signed geometric progressions with ratios $4^{-1}, 4^{-2}, \dots, 4^{-M}$. The proof is not an appeal to a merely asymptotic Fourier expansion. It uses two finite mechanisms: the omitted tail is a scaled copy of up , and for $n \geq d$ its support lies exactly in one polynomial spline cell centered at x ; moreover, every derivative $\text{up}^{(r)}(x)$ with $r > d$ vanishes.

Putting $Q = 1/4$, geometric Lagrange interpolation at the nodes $1, Q, \dots, Q^M$ gives the requested one-line extraction formula

$$\text{up}(x) = \frac{1}{(Q; Q)_M} \sum_{j=0}^M (-1)^{M-j} Q^{\binom{M-j+1}{2}} \begin{bmatrix} M \\ j \end{bmatrix}_Q \text{up}_{n+j}(x), \quad n \geq d.$$

All quantities on the right are rational. Hence this formula simultaneously extracts the limit exactly and gives a direct finite proof of the rationality of $\text{up}(x)$ at dyadic points. We also give the complete q -Vandermonde system for all geometric coefficients, an explicit inverse formula, a finite q -binomial annihilating recurrence, stability bounds, worked rational examples, and exact-arithmetic Python verification.

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1 Introduction

The Rvachev up-function is a compactly supported, even, infinitely differentiable function whose Fourier transform is an infinite dyadic sinc product. It may equally be viewed as the probability density of an infinite sum of independent centered uniform random variables. Truncating the Fourier product produces a compactly supported box spline. The resulting finite approximants are especially rigid: their knots form a uniform dyadic grid and the jumps of the top derivative carry the Thue–Morse signs.

The documentation in the `ProveIt` repository develops these finite sinc products, their exact truncated-power representation, the self-similar head–tail factorization, a uniform all-orders differential expansion, and geometric Richardson filters with closed Gaussian q -binomial weights [7]. Those ingredients suggest the following sharper pointwise question.

For a fixed dyadic rational x , does the asymptotic series in powers of 4^{-n} become an *exact finite formula*, and can the true value $\text{up}(x)$ be recovered from finitely many rational spline values by one closed expression?

The answer is yes. The decisive observation is geometric. Once the spline level reaches the denominator depth of x , the point x is the midpoint of a spline cell, and the entire support of the omitted scaled tail is exactly that cell. Convolution by the tail therefore acts on one finite-dimensional polynomial space. Its inverse is not an infinite pseudodifferential series on that space: it is a finite nilpotent differential operator. The coefficients of this inverse are precisely the reciprocal-product coefficients already visible in Fourier analysis. A second finite phenomenon then occurs: the Taylor jet of up at a dyadic point terminates. Combining the two gives an exact sum of geometric modes.

The limit is the constant coefficient of a polynomial sampled at the geometric nodes $1, Q, \dots, Q^M$, where $Q = 1/4$. Evaluating that polynomial at zero is elementary Lagrange interpolation, and simplification of the Lagrange weights produces the advertised q -Pochhammer/Gaussian-binomial formula. This is also a particularly well-conditioned extrapolation: the total variation of the weights stays below 1.97, uniformly in the dyadic depth.

Main conclusions

For a dyadic point of depth d , set $M = \lfloor d/2 \rfloor$. Then:

1. the guaranteed transient ends at level $n = d$;
2. $q_n = \text{up}_n(x)$ is exactly a polynomial of degree at most M in Q^{n+1} , and the degree is M for every nonintegral reduced dyadic point;
3. $M + 1$ consecutive rational values determine every mode coefficient;
4. the constant term $\text{up}(x)$ is obtained by a single universal q -binomial filter depending only on d , not on the numerator of x ;
5. the same sequence satisfies an exact order- $M + 1$ recurrence whose coefficients are Gaussian q -binomials.

All numerical calculations in the accompanying script use `fractions.Fraction`. They are verification and illustration only; every mathematical assertion used by the extraction method is proved below.

2 Fourier normalization and the finite box splines

2.1 The infinite product

We use the Fourier convention

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-2\pi i t x} dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}(t) e^{2\pi i t x} dt. \quad (1)$$

For $a > 0$, let

$$b_a(x) = \frac{1}{2a} \mathbf{1}_{[-a, a]}(x).$$

Then

$$\widehat{b}_a(t) = \text{sinc}(2\pi a t).$$

Consequently the normalized up-function is the probability density with Fourier transform

$$\Phi(t) := \widehat{\text{up}}(t) = \prod_{k=0}^{\infty} \text{sinc}\left(\frac{\pi t}{2^k}\right), \quad (2)$$

and support $[-1, 1]$. This is the normalization used in the classical work of Rvachev and Arias de Reyna [1, 2].

For $n \geq 0$, define the user-indexed prefix

$$\Phi_n(t) = \prod_{k=0}^n \text{sinc}\left(\frac{\pi t}{2^k}\right), \quad \widehat{\text{up}}_n = \Phi_n. \quad (3)$$

There are $n + 1$ factors, so

$$\text{up}_n = b_{2^{-1}} * b_{2^{-2}} * \cdots * b_{2^{-(n+1)}}. \quad (4)$$

In particular, up_n is a probability density and is a polynomial spline of degree n .

2.2 Random-sum and exact tail factorization

Let $U_0, U_1, \dots$ be independent random variables, each uniform on $[-1, 1]$, and put

$$S_n = \sum_{k=0}^n 2^{-k-1} U_k, \quad X = \sum_{k=0}^{\infty} 2^{-k-1} U_k. \quad (5)$$

Then up_n and up are the densities of S_n and X , respectively. The omitted tail is

$$\sum_{k=n+1}^{\infty} 2^{-k-1} U_k = 2^{-(n+1)} \sum_{j=0}^{\infty} 2^{-j-1} U_{n+1+j}.$$

Thus, with

$$\delta_n = 2^{-(n+1)}, \quad (6)$$

we have the distributional identity

$$X \stackrel{d}{=} S_n + \delta_n X', \quad (7)$$

where X' is an independent copy of X . At the density and Fourier levels this becomes

$$\Phi(t) = \Phi_n(t) \Phi(\delta_n t), \quad \text{up} = \text{up}_n * \rho_{\delta_n}, \quad \rho_{\delta}(y) = \delta^{-1} \text{up}(y/\delta). \quad (8)$$

The support of ρ_{δ} is $[-\delta, \delta]$.

3 Exact spline arithmetic and dyadic cell geometry

3.1 Thue–Morse truncated-power formula

Let

$$\tau_k = (-1)^{s_2(k)},$$

where $s_2(k)$ is the sum of the binary digits of k . Repeatedly differentiating the boxes in Equation (4), or integrating the resulting atomic top derivative, gives the following exact formula.

Proposition 3.1 (Exact finite-spline formula). *For every $n \geq 0$, away from the knots when $n = 0$,*

$$\text{up}_n(x) = \frac{2^{n(n+1)/2}}{n!} \sum_{k=0}^{2^{n+1}-1} \tau_k \left(x + 1 - 2^{-(n+1)} - \frac{k}{2^n} \right)_+^n. \quad (9)$$

Its knots are

$$t_{n,k} = -1 + 2^{-(n+1)} + \frac{k}{2^n} = -1 + (2k+1)2^{-(n+1)}, \quad 0 \leq k < 2^{n+1}. \quad (10)$$

The support is $[-1 + 2^{-(n+1)}, 1 - 2^{-(n+1)}]$, and on each open knot cell up_n is a polynomial of degree at most n .

Proof. For a centered box,

$$Db_a = \frac{1}{2a}(\delta_{-a} - \delta_a).$$

Taking one derivative from every factor in Equation (4) shows that $D^{n+1}\text{up}_n$ is a signed sum of 2^{n+1} point masses. Choosing right endpoints according to a binary vector gives the sign τ_k , the uniform location in Equation (10), and the normalization

$$\prod_{j=0}^n 2^j = 2^{n(n+1)/2}.$$

Integrating $n+1$ times from $-\infty$, using compact support, yields Equation (9). The support and polynomial-cell statements are immediate. $\square$

Corollary 3.2 (Rational prefix values). *If $x \in \mathbb{Q}$, then $\text{up}_n(x) \in \mathbb{Q}$. In particular every term $q_n = \text{up}_n(x)$ is rational when x is dyadic.*

3.2 Dyadic depth and the centered-cell lemma

Definition 3.3 (Dyadic depth). *For a dyadic rational x , define*

$$d(x) = \min\{d \geq 0 : 2^d x \in \mathbb{Z}\}.$$

If $x \notin \mathbb{Z}$, this says that $x = a/2^d$ in lowest terms with a odd.

The next elementary observation is the geometric reason for the exact termination theorem.

Lemma 3.4 (A dyadic point becomes a cell midpoint). *Let $x \in (-1, 1)$ be dyadic of depth d . For every $n \geq d$, the two knots adjacent to x in the spline up_n are*

$$x - \delta_n \quad \text{and} \quad x + \delta_n, \quad \delta_n = 2^{-(n+1)}.$$

Hence $(x - \delta_n, x + \delta_n)$ is one polynomial cell of up_n , centered at x , and it is exactly as wide as the support of the omitted tail kernel in Equation (8).

Proof. Since $n \geq d$, the number

$$J = 2^n(x + 1)$$

is an integer. Therefore

$$x = -1 + \frac{J}{2^n} = -1 + 2J2^{-(n+1)}.$$

By Equation (10), the adjacent odd-grid points are

$$-1 + (2J - 1)2^{-(n+1)} = x - \delta_n, \quad -1 + (2J + 1)2^{-(n+1)} = x + \delta_n.$$

There is no knot strictly between them. $\square$

Remark 3.5. *The guaranteed threshold in the user's indexing is therefore exactly $n = d$. Some special points may satisfy the final formula at earlier levels, but no earlier universal threshold follows from the cell geometry.*

4 Reciprocal-product coefficients and finite local deconvolution

4.1 Moments and the reciprocal sinc product

Let

$$\mu_r = \int_{-1}^1 t^r \text{up}(t) dt = \mathbb{E}[X^r]. \quad (11)$$

The density is even, so $\mu_{2r+1} = 0$. Its moment-generating series is

$$B(s) = \mathbb{E}[e^{sX}] = \sum_{m=0}^{\infty} \frac{\mu_{2m}}{(2m)!} s^{2m}. \quad (12)$$

Because $\Phi(z) = B(-2\pi iz)$, define a_m near the origin by

$$\frac{1}{\Phi(z)} = \sum_{m=0}^{\infty} a_m (2\pi z)^{2m}. \quad (13)$$

Substituting $z = is/(2\pi)$ gives the inverse moment series

$$\frac{1}{B(s)} = \sum_{m=0}^{\infty} (-1)^m a_m s^{2m}. \quad (14)$$

Equivalently, if $b_m = \mu_{2m}/(2m)!$, then

$$\sum_{j=0}^m b_j (-1)^{m-j} a_{m-j} = \delta_{m0}. \quad (15)$$

The logarithm of the sinc product gives an explicit rational description. Put $Q = 1/4$. Then

$$\frac{1}{\Phi(z)} = \exp\left(\sum_{j=1}^{\infty} \alpha_j (2\pi z)^{2j}\right), \quad \alpha_j = \frac{|B_{2j}|}{2j(2j)!(1-Q^j)}, \quad (16)$$

where B_{2j} are the Bernoulli numbers. Hence

$$a_0 = 1, \quad ma_m = \sum_{j=1}^m j \alpha_j a_{m-j}, \quad (17)$$

and, using the complete exponential Bell polynomial Y_m ,

$$a_m = \frac{1}{m!} Y_m(1!\alpha_1, 2!\alpha_2, \dots, m!\alpha_m). \quad (18)$$

In particular, every a_m is a positive rational number. The first values are

$$a_0 = 1, \quad a_1 = \frac{1}{18}, \quad a_2 = \frac{31}{16200}, \quad a_3 = \frac{2347}{42865200}, \quad a_4 = \frac{1904369}{1311675120000}. \quad (19)$$

4.2 An exact local inverse, not an asymptotic one

For an arbitrary point, formally dividing by $\Phi(\delta_n t)$ leads to an all-orders asymptotic differential expansion. At the centered dyadic cell the same calculation is exact because it takes place on a finite-dimensional polynomial space.

Theorem 4.1 (Finite local deconvolution). *Let $x \in (-1, 1)$ be dyadic of depth d , let $n \geq d$, and let $0 \leq r \leq n$. Then*

$$\boxed{\text{up}_n^{(r)}(x) = \sum_{m=0}^{\lfloor (n-r)/2 \rfloor} (-1)^m a_m \delta_n^{2m} \text{up}^{(r+2m)}(x), \quad \delta_n = 2^{-(n+1)}. \quad (20)}$$

This identity is exact for each such n ; there is no remainder term.

Proof. By [Theorem 3.4](#), up_n agrees on $(x - \delta_n, x + \delta_n)$ with one polynomial g of degree at most n . From the exact convolution identity [Equation \(8\)](#), differentiation in the sense of distributions followed by evaluation at x gives, for $0 \leq r \leq n$,

$$\begin{aligned} \text{up}_n^{(r)}(x) &= \int_{-1}^1 \text{up}_n^{(r)}(x - \delta_n t) \text{up}(t) dt \\ &= \int_{-1}^1 g^{(r)}(x - \delta_n t) \text{up}(t) dt. \end{aligned} \quad (21)$$

The possible cell endpoints correspond to $t = \pm 1$ and have measure zero. Since $g^{(r)}$ is a polynomial, its Taylor expansion is finite. Odd moments vanish, so

$$\text{up}_n^{(r)}(x) = \sum_{j=0}^{\lfloor (n-r)/2 \rfloor} b_j \delta_n^{2j} \text{up}_n^{(r+2j)}(x), \quad b_j = \frac{\mu_{2j}}{(2j)!}. \quad (22)$$

This is a unit upper-triangular relation between the even subjets. Its inverse is determined by the reciprocal power series of $\sum b_j s^{2j} = B(s)$. By [Equation \(14\)](#), the inverse coefficients are $(-1)^m a_m$. Because the derivative shift is nilpotent on polynomials of degree at most n , multiplying the two finite triangular sums uses only the finite coefficient identities [Equation \(15\)](#); no convergence argument is involved. Applying the inverse relation to [Equation \(22\)](#) proves [Equation \(20\)](#). $\square$

Remark 4.2 (Operator interpretation). *On the local polynomial g , convolution by the tail is the finite operator*

$$B(-\delta_n D)g = B(\delta_n D)g,$$

because B is even. Its inverse is

$$B(\delta_n D)^{-1} = \sum_{m \geq 0} (-1)^m a_m \delta_n^{2m} D^{2m},$$

and the sum terminates automatically since $D^{n+1}g = 0$. This is the local, finite-dimensional meaning of division by the omitted Fourier factor.

5 Termination of the dyadic Taylor jet

We next prove the second finite mechanism: sufficiently high derivatives of up vanish at every dyadic point.

5.1 Functional-differential equation

The product identity

$$\Phi(t) = \text{sinc}(\pi t) \Phi(t/2)$$

is equivalent, under Equation (1), to

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1). \quad (23)$$

Iterating this relation gives the standard Thue–Morse derivative formula.

Lemma 5.1 (Iterated derivative identity). *For every integer $r \geq 0$,*

$$\boxed{\text{up}^{(r)}(x) = 2^{r(r+1)/2} \sum_{k=0}^{2^r-1} \tau_k \text{up}(2^r x + 2^r - 1 - 2k).} \quad (24)$$

Proof. The case $r = 0$ is tautological, and the case $r = 1$ is Equation (23). Suppose the formula holds for r . Differentiating introduces a factor 2^r , and applying Equation (23) to each summand introduces another factor 2 and two children whose signs are related by

$$\tau_{2k} = \tau_k, \quad \tau_{2k+1} = -\tau_k.$$

The total new factor is 2^{r+1} , changing $2^{r(r+1)/2}$ into $2^{(r+1)(r+2)/2}$, and the children enumerate $0 \leq k < 2^{r+1}$. This proves the induction step. $\square$

5.2 Dyadic flatness

Theorem 5.2 (Dyadic jet termination). *Let $x \in [-1, 1]$ be dyadic of depth d . Then*

$$\text{up}^{(r)}(x) = 0 \quad \text{for every } r > d. \quad (25)$$

For a nonintegral reduced dyadic point, the highest even derivative permitted by this bound is nonzero: if $M = \lfloor d/2 \rfloor$, then

$$\text{up}^{(2M)}(x) \neq 0. \quad (26)$$

Proof. First take $r > d$. Then $2^r x$ is an even integer. Every shift $2^r - 1 - 2k$ in Equation (24) is odd, so every argument of up in that formula is an odd integer. Since up is supported on $[-1, 1]$ and is smooth, it vanishes at ± 1 and outside the support. Thus every summand is zero, proving Equation (25). The same argument includes $x = 0, \pm 1$.

It remains to show Equation (26) for a nonintegral point $x = a/2^d$ in lowest terms. We first record the elementary values

$$\text{up}(0) = 1, \quad \text{up}(1/2) = \text{up}(-1/2) = 1/2. \quad (27)$$

Indeed, for $0 \leq y \leq 1$, differentiating $h(y) = \text{up}(y) + \text{up}(y - 1)$ and using Equation (23) shows that h is constant; its integral over $[0, 1]$ is the total mass 1, so $h \equiv 1$. At $y = 0$, endpoint flatness gives $\text{up}(0) = 1$; at $y = 1/2$, evenness gives $2 \text{up}(1/2) = 1$.

If $d = 2M$ is even, put $r = d$ in Equation (24). The arguments are even integers spaced by two, and exactly one is zero. Therefore

$$\text{up}^{(d)}(x) = \pm 2^{d(d+1)/2} \text{up}(0) \neq 0.$$

If $d = 2M + 1$ is odd, put $r = d - 1 = 2M$. The arguments are half-integers spaced by two, and exactly one of them is $1/2$ or $-1/2$; all others lie outside $[-1, 1]$. Hence

$$\text{up}^{(d-1)}(x) = \pm 2^{(d-1)d/2} \text{up}(1/2) \neq 0.$$

This is precisely Equation (26). □

Corollary 5.3 (Terminating Taylor polynomial). *At a dyadic point of depth d , the formal Taylor series of up contains no term of degree greater than d . This finite Taylor polynomial need not represent up on any neighborhood; indeed the up-function is nowhere analytic on its support [2].*

6 The exact eventual geometric formula

We can now combine the centered-cell inverse with dyadic jet termination.

Theorem 6.1 (Exact finite geometric expansion). *Fix a dyadic rational $x \in [-1, 1]$, let*

$$d = d(x), \quad M = \left\lfloor \frac{d}{2} \right\rfloor, \quad Q = \frac{1}{4},$$

and set $q_n = \text{up}_n(x)$. Then, for every $n \geq d$,

$$q_n = \sum_{m=0}^M (-1)^m a_m \text{up}^{(2m)}(x) Q^{m(n+1)}. \quad (28)$$

Equivalently,

$$q_n = \text{up}(x) + \sum_{m=1}^M C_m(x) Q^{m(n+1)}, \quad C_m(x) = (-1)^m a_m \text{up}^{(2m)}(x). \quad (29)$$

Thus the nonconstant terms are signed geometric progressions whose successive ratios are

$$Q^m = 4^{-m}, \quad 1 \leq m \leq M.$$

For a nonintegral reduced dyadic point, the coefficient of the highest mode $Q^{M(n+1)}$ is nonzero, so the geometric polynomial has degree exactly M .

Proof. For $x \in (-1, 1)$ and $n \geq d$, apply Theorem 4.1 with $r = 0$. Its finite sum initially runs to $\lfloor n/2 \rfloor$. By Theorem 5.2, every derivative with order greater than d vanishes, so only $2m \leq d$, that is $m \leq M$, survive. Since $\delta_n^{2m} = 4^{-m(n+1)} = Q^{m(n+1)}$, this is Equation (28). The endpoint sequences are identically zero, and the central point sequence is identically one, so the same statement holds at $x = -1, 0, 1$. Finally, $a_M > 0$ and $\text{up}^{(2M)}(x) \neq 0$ by Theorem 5.2, proving exact degree for nonintegral reduced points. □

Remark 6.2 (Why the formula is stronger than the global expansion). *Fourier division gives, for fixed truncation order K , an asymptotic formula with a remainder $O(4^{-Kn})$ uniformly in x . At a dyadic point, the present proof does not attempt to make that remainder small. It removes it: local convolution is inverted exactly on one spline polynomial, and all remaining derivatives above the dyadic depth are exactly zero.*

Corollary 6.3 (Exact rationality from finite splines). *For every dyadic $x \in [-1, 1]$, $\text{up}(x) \in \mathbb{Q}$. Moreover all mode coefficients $C_m(x)$ are rational.*

Proof. The next section expresses $\text{up}(x)$ and every $C_m(x)$ as rational linear combinations of finitely many rational prefix values from [Theorem 3.2](#). This gives a finite proof independent of taking a limit of rationals. Rationality of all dyadic derivatives then also follows from [Equation \(24\)](#). $\square$

7 Geometric interpolation and the one-line extraction formula

7.1 The q -Vandermonde system

Fix a starting level $n_0 \geq d$. Rewrite [Equation \(29\)](#) as

$$q_{n_0+j} = \sum_{m=0}^M B_m Q^{jm}, \quad B_m = C_m(x) Q^{m(n_0+1)}, \quad 0 \leq j \leq M. \quad (30)$$

Thus $(B_0, \dots, B_M)^T$ is the solution of the exact system

$$\begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & Q & Q^2 & \cdots & Q^M \\ 1 & Q^2 & Q^4 & \cdots & Q^{2M} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & Q^M & Q^{2M} & \cdots & Q^{M^2} \end{pmatrix} \begin{pmatrix} B_0 \\ B_1 \\ \vdots \\ B_M \end{pmatrix} = \begin{pmatrix} q_{n_0} \\ q_{n_0+1} \\ \vdots \\ q_{n_0+M} \end{pmatrix}. \quad (31)$$

Its determinant is

$$\prod_{0 \leq r < s \leq M} (Q^s - Q^r) = (-1)^{\binom{M+1}{2}} Q^{\binom{M+1}{3}} \prod_{s=1}^M (Q; Q)_s, \quad (32)$$

which is nonzero because $0 < Q < 1$. Hence $M+1$ consecutive values uniquely determine every geometric coefficient. The desired limit is simply

$$B_0 = C_0(x) = \text{up}(x).$$

7.2 Lagrange evaluation at zero

Let

$$P(z) = \sum_{m=0}^M B_m z^m.$$

Then $P(Q^j) = q_{n_0+j}$, and $\text{up}(x) = P(0)$. Lagrange interpolation gives

$$P(0) = \sum_{j=0}^M W_{M,j} q_{n_0+j}, \quad W_{M,j} = \prod_{\substack{0 \leq \ell \leq M \\ \ell \neq j}} \frac{-Q^\ell}{Q^j - Q^\ell}. \quad (33)$$

We now simplify the product.

For $\ell < j$,

$$\prod_{\ell=0}^{j-1} (Q^j - Q^\ell) = (-1)^j Q^{j(j-1)/2} (Q; Q)_j,$$

while for $\ell > j$,

$$\prod_{\ell=j+1}^M (Q^j - Q^\ell) = Q^{j(M-j)} (Q; Q)_{M-j}.$$

The numerator in Equation (33) is $(-1)^M Q^{M(M+1)/2-j}$. Cancelling powers and signs yields

$$W_{M,j} = (-1)^{M-j} \frac{Q^{\binom{M-j+1}{2}}}{(Q; Q)_j (Q; Q)_{M-j}}. \quad (34)$$

Using

$$\begin{bmatrix} M \\ j \end{bmatrix}_Q = \frac{(Q; Q)_M}{(Q; Q)_j (Q; Q)_{M-j}}, \quad (35)$$

we obtain the central extraction theorem.

Theorem 7.1 (Exact q -binomial extraction). *Let $x \in [-1, 1]$ be dyadic of depth d , put $M = \lfloor d/2 \rfloor$ and $Q = 1/4$, and choose any $n \geq d$. Then*

$$\text{up}(x) = \sum_{j=0}^M (-1)^{M-j} \frac{Q^{\binom{M-j+1}{2}}}{(Q; Q)_j (Q; Q)_{M-j}} q_{n+j}. \quad (36)$$

Equivalently, in the requested single Gaussian-binomial/ q -Pochhammer form,

$$\text{up}(x) = \frac{1}{(Q; Q)_M} \sum_{j=0}^M (-1)^{M-j} Q^{\binom{M-j+1}{2}} \begin{bmatrix} M \\ j \end{bmatrix}_Q \text{up}_{n+j}(x), \quad Q = \frac{1}{4}, \quad n \geq d. \quad (37)$$

Proof. The exact geometric formula makes q_{n+j} the values of a polynomial of degree at most M at Q^j . Lagrange interpolation at zero proves Equation (33), and the preceding product calculation proves Equations (36) and (37). $\square$

7.3 Shift-operator form

Let $\mathbf{E}q_n = q_{n+1}$. Every nonconstant mode $Q^{m(n+1)}$ is an eigenvector of $\mathbf{E}$ with eigenvalue Q^m , while the constant mode has eigenvalue 1. Therefore

$$\text{up}(x) = \frac{(\mathbf{E} - Q)(\mathbf{E} - Q^2) \cdots (\mathbf{E} - Q^M)}{(1 - Q)(1 - Q^2) \cdots (1 - Q^M)} q_n, \quad n \geq d. \quad (38)$$

The denominator is $(Q; Q)_M$. Expanding the numerator by the finite q -binomial theorem gives exactly Equation (37). This form makes the mechanism transparent: the filter preserves the constant mode and kills every permitted decaying mode.

7.4 First universal filters

For $Q = 1/4$, the first cases are

$$M = 0 : \quad \text{up}(x) = q_n, \quad (39)$$

$$M = 1 : \quad \text{up}(x) = \frac{4q_{n+1} - q_n}{3}, \quad (40)$$

$$M = 2 : \quad \text{up}(x) = \frac{q_n - 20q_{n+1} + 64q_{n+2}}{45}, \quad (41)$$

$$M = 3 : \quad \text{up}(x) = \frac{-q_n + 84q_{n+1} - 1344q_{n+2} + 4096q_{n+3}}{2835}. \quad (42)$$

The same filter applies to every numerator at a given denominator-depth band: $M = 1$ for depths 2, 3, $M = 2$ for depths 4, 5, and so forth.

8 Recovery of every geometric coefficient

The limit requires only the first row of the inverse Vandermonde matrix. For completeness, this section gives all rows explicitly.

Let

$$D_j = \prod_{\substack{0 \leq \ell \leq M \\ \ell \neq j}} (Q^j - Q^\ell) = (-1)^j Q^{j(j-1)/2 + j(M-j)} (Q; Q)_j (Q; Q)_{M-j}. \quad (43)$$

Define the elementary symmetric quantity

$$e_r^{(j)} = e_r(1, Q, \dots, \widehat{Q^j}, \dots, Q^M) = [t^r] \frac{(-t; Q)_{M+1}}{1 + tQ^j}, \quad (44)$$

where the hat means omission and $[t^r]$ extracts a coefficient. The finite q -binomial theorem also gives

$$e_r^{(j)} = \sum_{h=0}^r (-1)^h Q^{jh + \binom{r-h}{2}} \begin{bmatrix} M+1 \\ r-h \end{bmatrix}_Q. \quad (45)$$

Proposition 8.1 (Closed inverse for all modes). *With the notation of Equation (30),*

$$B_m = \sum_{j=0}^M q_{n_0+j} \frac{(-1)^{M-m} e_{M-m}^{(j)}}{D_j}, \quad 0 \leq m \leq M, \quad (46)$$

and hence

$$C_m(x) = Q^{-m(n_0+1)} \sum_{j=0}^M q_{n_0+j} \frac{(-1)^{M-m} e_{M-m}^{(j)}}{D_j}. \quad (47)$$

For $m = 0$, this reduces to Equation (36).

Proof. The Lagrange basis polynomial for the node Q^j is

$$L_j(z) = \frac{\prod_{\ell \neq j} (z - Q^\ell)}{D_j}.$$

Expanding its numerator in elementary symmetric functions gives

$$[z^m] L_j(z) = \frac{(-1)^{M-m} e_{M-m}^{(j)}}{D_j}.$$

Since $P(z) = \sum_j q_{n_0+j} L_j(z)$, coefficient comparison proves Equation (46); the rescaling in Equation (30) proves Equation (47). Formula Equation (45) follows by expanding $(-t; Q)_{M+1}$ and multiplying by the geometric series for $(1 + tQ^j)^{-1}$, truncated at degree r . $\square$

In practice, exact Gaussian elimination on Equation (31) is often simpler than evaluating Equation (47). The proposition is useful conceptually: it proves that every derivative combination $(-1)^m a_m \text{up}^{(2m)}(x)$ is a rational expression in finitely many rational spline values.

9 An exact q -binomial recurrence and model diagnostics

Because q_n is a linear combination of the modes with eigenvalues $1, Q, \dots, Q^M$, it is annihilated by their characteristic polynomial.

Theorem 9.1 (Eventual annihilating recurrence). *For every $n \geq d$,*

$$\prod_{m=0}^M (\mathbf{E} - Q^m) q_n = 0. \quad (48)$$

Equivalently,

$$\sum_{j=0}^{M+1} (-1)^{M+1-j} Q^{\binom{M+1-j}{2}} \begin{bmatrix} M+1 \\ j \end{bmatrix}_Q q_{n+j} = 0. \quad (49)$$

Proof. Each factor $\mathbf{E} - Q^m$ kills the corresponding geometric mode in Equation (28). Expanding

$$\prod_{m=0}^M (z - Q^m) = z^{M+1} (z^{-1}; Q)_{M+1}$$

by the finite q -binomial theorem yields Equation (49). $\square$

This recurrence provides three useful diagnostics.

1. **Threshold detection.** If one computes exact prefix values without trusting the theoretical threshold, the first level from which Equation (49) holds on several successive blocks identifies the onset of the geometric regime. The theorem guarantees onset no later than $n = d$.
2. **Model-order verification.** Testing orders $0, 1, \dots, M$ reveals whether accidental lower-mode cancellations reduce the effective model. For a reduced nonintegral dyadic point the degree is exactly M , so order M is necessary.
3. **Error detection.** In exact rational arithmetic, a nonzero residual is an unambiguous implementation error or an indication that the starting index is still in the transient range.

10 Stability of the extraction filter

Although the weights alternate in sign, they remain uniformly tame. From Equation (34),

$$\sum_{j=0}^M |W_{M,j}| = \frac{\prod_{r=1}^M (1 + Q^r)}{\prod_{r=1}^M (1 - Q^r)} = \frac{(-Q; Q)_M}{(Q; Q)_M}. \quad (50)$$

For $Q = 1/4$, this increases to the finite limit

$$\frac{(-Q; Q)_\infty}{(Q; Q)_\infty} = 1.969260353668 \dots \quad (51)$$

Consequently, if approximate input values have absolute errors at most ε , the extracted value has absolute error at most

$$1.970 \varepsilon.$$

There is no depth-dependent explosion of the extrapolation coefficients. Of course, when exact rational prefix values are supplied, the extraction itself is exact.

The bounded norm is unusually favorable for high-order extrapolation. It is a consequence of the strongly separated geometric nodes $1, 1/4, 1/16, \dots$, not a generic property of polynomial extrapolation.

11 Worked exact examples

The following examples use the user's indexing: up_n contains the factors $k = 0, \dots, n$.

11.1 Depth two: $x = 1/4$

Here $d = 2$, $M = 1$, and the geometric formula begins at $n = 2$. Exact spline evaluation gives

$$q_2 = \frac{15}{16}, \quad q_3 = \frac{179}{192}.$$

The universal two-level filter [Equation \(40\)](#) yields

$$\text{up}\left(\frac{1}{4}\right) = \frac{4q_3 - q_2}{3} = \frac{67}{72}.$$

Solving the two-by-two system gives the complete sequence

$$\boxed{\text{up}_n\left(\frac{1}{4}\right) = \frac{67}{72} + \frac{4}{9} 4^{-(n+1)}, \quad n \geq 2.} \quad (52)$$

The mode coefficient agrees with the differential formula because $a_1 = 1/18$ and $\text{up}''(1/4) = -8$.

11.2 Depth three: $x = 1/8$

Here $d = 3$, but still $M = 1$. The first guaranteed samples are

$$q_3 = \frac{383}{384}, \quad q_4 = \frac{1531}{1536}.$$

Therefore

$$\text{up}\left(\frac{1}{8}\right) = \frac{4q_4 - q_3}{3} = \frac{287}{288},$$

and the full eventual formula is

$$\boxed{\text{up}_n\left(\frac{1}{8}\right) = \frac{287}{288} + \frac{2}{9} 4^{-(n+1)}, \quad n \geq 3.} \quad (53)$$

11.3 Depth four: $x = 1/16$

Now $d = 4$ and $M = 2$, so two decaying modes occur. The first three exact values in the guaranteed regime are

$$\begin{aligned} q_4 &= \frac{24575}{24576}, \\ q_5 &= \frac{1965959}{1966080}, \\ q_6 &= \frac{94365509}{94371840}. \end{aligned}$$

Applying Equation (41),

$$\text{up}\left(\frac{1}{16}\right) = \frac{q_4 - 20q_5 + 64q_6}{45} = \frac{2073457}{2073600}. \quad (54)$$

Solving the complete Vandermonde system gives

$$\boxed{\text{up}_n\left(\frac{1}{16}\right) = \frac{2073457}{2073600} + \frac{5}{162} 4^{-(n+1)} - \frac{3968}{2025} 16^{-(n+1)}, \quad n \geq 4.} \quad (55)$$

From $a_1 = 1/18$ and $a_2 = 31/16200$, this corresponds to

$$\text{up}''\left(\frac{1}{16}\right) = -\frac{5}{9}, \quad \text{up}^{(4)}\left(\frac{1}{16}\right) = -1024.$$

11.4 Same depth, different numerator: $x = 3/16$

The extraction weights are unchanged because they depend only on d . The samples

$$q_4 = \frac{8011}{8192}, \quad q_5 = \frac{1922041}{1966080}, \quad q_6 = \frac{92250811}{94371840}$$

give

$$\text{up}\left(\frac{3}{16}\right) = \frac{2026943}{2073600} \quad (56)$$

and

$$\text{up}_n\left(\frac{3}{16}\right) = \frac{2026943}{2073600} + \frac{67}{162} 4^{-(n+1)} + \frac{3968}{2025} 16^{-(n+1)}, \quad n \geq 4. \quad (57)$$

The universal filter is the same; only the mode amplitudes change.

11.5 Degenerate low-depth cases

At the three integer points,

$$\text{up}_n(-1) = 0, \quad \text{up}_n(0) = 1, \quad \text{up}_n(1) = 0$$

for every $n \geq 0$. At $x = \pm 1/2$, $d = 1$ and $M = 0$, so

$$\text{up}_n(\pm 1/2) = \text{up}(\pm 1/2) = \frac{1}{2}, \quad n \geq 1.$$

These are exact stabilization rather than nontrivial extrapolation.

12 A practical exact algorithm

12.1 Input and guaranteed sample count

Given a dyadic rational $x \in [-1, 1]$:

1. Reduce x to lowest terms and determine $d = \min\{r : 2^r x \in \mathbb{Z}\}$.
2. Set $M = \lfloor d/2 \rfloor$ and $Q = 1/4$.
3. Choose any starting level $n_0 \geq d$; the smallest guaranteed choice is $n_0 = d$.
4. Compute the $M + 1$ exact rational values $q_{n_0}, q_{n_0+1}, \dots, q_{n_0+M}$, for example from [Equation \(9\)](#).
5. Return the dot product in [Equation \(37\)](#).

This returns $\text{up}(x)$ exactly. No limiting process, numerical tolerance, or rational reconstruction is needed.

12.2 Recovering and certifying the complete sequence

To obtain every coefficient, solve [Equation \(31\)](#) exactly and undo the scale:

$$C_m = Q^{-m(n_0+1)} B_m.$$

Then certify the result in either of two ways:

1. compare the predicted formula with one or more additional exact values $q_{n_0+M+1}, q_{n_0+M+2}, \dots$; or
2. evaluate the exact recurrence residual in [Equation \(49\)](#).

The proof already guarantees correctness, but these checks are valuable for software regression testing.

12.3 Complexity

The direct truncated-power sum uses $O(2^{n+1})$ terms. Since the minimal sample block reaches roughly $n = d + \lfloor d/2 \rfloor$, this transparent method is exponential in the dyadic depth. The extrapolation itself is cheap: weight construction takes $O(M^2)$ rational operations if powers and q -Pochhammers are accumulated, and a full dense solve takes $O(M^3)$. For deep points, the prefix values can be accelerated by recursive Thue–Morse prefix-power sums, repeated-integration recurrences, or existing exact dyadic algorithms [3, 4]. The extraction formula is independent of how the input prefix values are obtained.

13 Exact-arithmetic experiments

The archive accompanying this article contains

`verify_dyadic_extraction.py`

and its captured output. The script implements:

1. the exact truncated-power spline formula with Thue–Morse signs;

2. $(Q; Q)_r$, Gaussian q -binomials, and the extraction weights;
3. exact Gauss–Jordan solution of the q -Vandermonde system;
4. prediction of unused spline levels;
5. the annihilating recurrence [Equation \(49\)](#);
6. an exhaustive regression over every reduced dyadic point of depth at most six.

Every calculation uses Python’s `Fraction`; a statement such as “residual zero” therefore means exact equality of integers after reduction, not agreement to a floating-point tolerance.

Representative output reproduces the examples in the preceding section. The exhaustive test checks 129 points, including the three integers, and finds that every held-out geometric prediction and every recurrence residual is exactly zero. This computation is ancillary: it tests the formulas and their indexing, while the proofs establish them for all dyadic depths.

14 Further consequences and extensions

14.1 Derivative extraction

The local theorem contains more than values. If $0 \leq r \leq d$, then for $n \geq d$,

$$\text{up}_n^{(r)}(x) = \sum_{m=0}^{\lfloor (d-r)/2 \rfloor} (-1)^m a_m \text{up}^{(r+2m)}(x) Q^{m(n+1)}. \quad (58)$$

Thus the same q -binomial filter, with $M_r = \lfloor (d-r)/2 \rfloor$, extracts $\text{up}^{(r)}(x)$ exactly from $M_r + 1$ consecutive derivative approximants. If $r > d$, both the true derivative and the local spline derivative vanish once the centered-cell regime is reached.

14.2 Using a larger stride

Suppose samples are available at levels $n, n+s, n+2s, \dots, n+Ms$. The corresponding interpolation nodes are

$$1, Q^s, Q^{2s}, \dots, Q^{Ms}.$$

Replacing Q by Q^s everywhere in [Equation \(37\)](#) gives another exact formula. This can be useful when only every s -th prefix has been computed.

14.3 Arbitrary selected levels

For distinct offsets $r_0, \dots, r_M$, ordinary Lagrange interpolation at nodes Q^{r_j} gives

$$\text{up}(x) = \sum_{j=0}^M q_{n+r_j} \prod_{\ell \neq j} \frac{-Q^{r_\ell}}{Q^{r_j} - Q^{r_\ell}}. \quad (59)$$

The consecutive case is distinguished because its products collapse to q -Pochhammer and Gaussian-binomial expressions.

14.4 A general local-deconvolution principle

The proof of [Theorem 4.1](#) is not special to sinc beyond the coefficient series. Let K be any compactly supported even probability density with moment-generating function B_K , and suppose a target density has an exact factorization

$$f = p_n * K_{\delta_n}.$$

If the support of K_{δ_n} , when centered at a point x , lies in one polynomial cell of p_n , then $p_n(x)$ is obtained from the finite jet of f by the reciprocal series $1/B_K$. If that jet terminates and δ_n^2 follows a geometric progression, the same exact geometric-mode and Lagrange-extraction theory follows. The Rvachev case is exceptionally clean because the tail is a scaled copy of the whole density and $\delta_n^2 = 4^{-(n+1)}$.

14.5 Formalization roadmap

The result is well suited to a modular formal proof. A possible decomposition is:

1. prove the uniform knot formula and the centered-cell lemma;
2. represent convolution with the scaled tail on the finite polynomial jet;
3. prove the reciprocal moment-series inversion in a nilpotent algebra;
4. formalize the Thue–Morse derivative identity and dyadic jet termination;
5. invoke the existing geometric Lagrange/ q -binomial infrastructure to derive [Equation \(37\)](#) and [Equation \(49\)](#).

The finite-dimensional inversion avoids analytic convergence issues at the main exactness step.

15 Conclusion

For a fixed dyadic rational, the finite-sinc-product approximants do not merely approach the Rvachev up-function with an all-orders expansion. After a precise and finite transient, the expansion terminates exactly. If d is the reduced dyadic depth, then every level $n \geq d$ satisfies

$$\text{up}_n(x) = \text{up}(x) + \sum_{m=1}^{\lfloor d/2 \rfloor} C_m(x) 4^{-m(n+1)}.$$

The reason is the conjunction of two exact geometries: the tail convolution is confined to one centered polynomial spline cell, and the up-function has a finite Taylor jet at the dyadic point.

The true rational value is therefore the value at zero of a degree $\lfloor d/2 \rfloor$ polynomial sampled on a geometric grid. Its Lagrange formula is the universal filter

$$\text{up}(x) = \frac{1}{(1/4; 1/4)_M} \sum_{j=0}^M (-1)^{M-j} 4^{-\binom{M-j+1}{2}} \left[\begin{matrix} M \\ j \end{matrix} \right]_{1/4} \text{up}_{n+j}(x), \quad M = \left\lfloor \frac{d}{2} \right\rfloor.$$

This formula uses only finitely many exact rational spline values, proves their limit is rational, recovers the limit without approximation, and fits directly into the q -binomial interpolation framework already developed around the Fabius–Rvachev system.

A A compact proof of the weight identities

For convenience, put

$$W_j = (-1)^{M-j} \frac{Q^{\binom{M-j+1}{2}}}{(Q; Q)_j (Q; Q)_{M-j}}.$$

The Lagrange argument proves

$$\sum_{j=0}^M W_j Q^{jr} = \delta_{r0}, \quad 0 \leq r \leq M. \quad (60)$$

One can also derive this directly from the finite q -binomial theorem. The shift polynomial is

$$\prod_{r=1}^M (z - Q^r) = \sum_{j=0}^M (-1)^{M-j} Q^{\binom{M-j+1}{2}} \begin{bmatrix} M \\ j \end{bmatrix}_Q z^j.$$

Dividing by its value at $z = 1$, namely $(Q; Q)_M$, gives the extraction filter. At $z = Q^r$, $1 \leq r \leq M$, it vanishes; at $z = 1$, it equals one. This is exactly Equation (60).

B Index conversion to the repository convention

Some source documents denote by p_N the inverse transform of the first N factors $k = 0, \dots, N-1$. The present article follows the user's notation up_n for factors $k = 0, \dots, n$. The conversion is

$$N = n + 1, \quad p_N = \text{up}_n, \quad 2^{-N} = 2^{-(n+1)} = \delta_n.$$

Thus a repository expansion in powers of 4^{-N} becomes the present expansion in powers of $4^{-(n+1)}$. Keeping this one-level shift explicit is important when reproducing the rational examples.

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